Module 2: Web Development

Deploy websites using frameworks and platforms.

Estimated Time: 3 hours

NOTE: When counting from 1 to 5, we go in order: 1, 2, 3, 4, 5. Similarly, when following a step-by-step process, it's important to go through each step in sequence without skipping around. If you find yourself stuck at any point, don't hesitate to reach out to your instructor for help and support.

If a text is in orange, then it is a link. A blue text should be typed **as is**. Then all red texts should be replaced by the expected value. Just ensure that you typed in **without misspellings**.

Learning Task 6 – Deploying Microservices

In older monolithic architectures, when one modules breaks, the whole system is unusable. Here we now call big software projects as systems, since they now involve the interaction of people, hardware, and software. In simple terms, the application is what the end-users interacts with, but this application is just part of a bigger and complex system. It is now an orchestra of software generating, managing, processing, receiving, and transmitting data across multiple client applications and backend servers.

With the popularity of microservices, different subsystems can now run in separate deployments and instances. It can also be used to strategically migrate from legacy systems to modern backend servers. The frontend web application often communicates through an API to transmit and receive data from the backend microservices. However, handling concurrency and data integrity in more complex systems would for another story.

Python

Although entry level software developers may eventually get promoted to senior roles, there are instances that due to the repetitive nature of work, many have their knowledge and skills get stagnated. Often times, corporate life in the IT industry also bombards software developers with tasks outside the typical *programming* as part of growing technically and contributing to company goals—this includes regular certifications, trainings, extension activities, event planning, and other side projects. Young software developers should avoid this trap of becoming obsolete by constantly learning new technologies outside their company tech stack no matter how busy corporate life gets.

When developing websites, developers are often siloed into tech stacks. This technological tribalism often arise from both academic background and industry experience. In practice, however, the choice of programming language and frameworks is often selected by the Chief Technology Officer

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(CTO), IT managers, and key senior software engineers. It would be a rare chance for a newbie to select the tech stack for a new software project.

Selecting a tech stack for large projects is not an easy task, it involves careful case studies to support long term success. It also factors unique client needs, current skills of existing staff, and long-term direction of a company. So, there is no hard rule that says a specific programming language is the "only solution" to all problems.

Over the years, Python have shown increasing popularity in web development. In selecting a framework, complexity of the software project must also be factored in. For example, Django is more appropriate for large projects which requires a full-stack and batteries-included framework for rapid web development. However, it can be an overkill for smaller software projects. Flask is an unopinionated framework used to build web applications with minimal setup. It is extensible and lightweight, which is great for microservices or if you just need the flexibility for customizations.

Before we continue, it is important to understand that the repositories that we deploy into cloud platforms are read-only. It means that we do not expect ourselves to upload files into repositories. Instead, backend websites acts as the mediator for the transmission of data from the frontend and into databases, filesystems, and other storage services. In this learning task, we will use Flask to handle our backend, but we will only use mock data and functions to simulate interaction of our frontend website and backend server.

Typically, the interaction of "human" actors with the use cases of the system are represented is use case diagram. This diagram is particularly helpful in the downstream system and analysis and design tasks. The breakdown of generic "human" actors is broken down in the table below.

Actors	User Level	Configure System Settings	Access Admin Menus	Access Basic Features	View Guest Features
Super administrator	3	✅	✅	✅	✅
Administrator	2	*	✅	*	✅
Moderator	1		*	*	✅
Standard user	0			✅	✅
Guest	-1			*	✅

* Access depends on the constraints of the system requirements.

These actors may have specific names depending on the actual software. For example, an administrator can be the "IT Personnel", a moderator can be the "Evaluator", and the standard user can be the "Contributor"—all of which are based on the context and nature of the software project being delivered. The **super administrator** often is a technically proficient individual that helps maintain the system, perform backup and recovery operations, and is able to access all use cases of a system. The **administrator** are assigned to set the preliminary data, customize the look & feel of a system, generate reports. The **moderator** is an optional actor which can help curate contents and prevent standard users

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from abusing the system. The **standard user** is a registered user which is the primary end-user that either *generate* contents or avail the services of the system. Lastly, the **guest** user is typically allowed to *browse* public contents, such as blogs, news, and e-commerce interface up to a certain degree.

The user level represents the hierarchy of different actors and it typically varies across different systems, but it requires that the lower the access level is, the closer it is to zero.

1. Go to the LT6 Dashboard spreadsheet.
    - Select another random repository which you haven't contributed yet.
        i. Negotiate with the owner to clarify and simplify the intent of the project.
    - After agreement, set your username as collaborator of the project.
        i. Ask to be a collaborator on **frontend** GitHub repository, if not yet added.
2. Go to GitHub (github.com) website.
    - Create a new repository named in "codename-api" format.
    - Add "**rmaniego**" and the *original project owner* (not you) as collaborator.
3. Go back to the **LT6 Dashboard** spreadsheet.
    - Paste the GitHub repository URL under the **backend server** column.
4. Go back to **Command Prompt**.
    - Clone the new repository locally.
    - Navigate **inside** the repository.
    - Create a ".gitignore" file: `type nul > .gitignore`
    - Open the ".gitignore" file and paste the following contents.

.gitignore

```
__pycache__
env/
```

    - Save your changes.
5. In **Command Prompt**, prepare the environment: (adjust commands on non-Windows platforms)

Command Prompt

```
pip install -U virtualenv
virtualenv env

echo .\env\Scripts\activate > activate.cmd
activate.cmd

pip install -U bcrypt flask flask_cors
pip freeze > requirements.txt
```

6. In the same **Command Prompt**, push the changes with the commit message:

Command Prompt

```
git add .gitignore
git commit -m "[env] Configure the dependencies."
git push -u origin main
```


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7. In the same **Command Prompt**.
   - Create the "data" folder.
   - Create the "data/mock-account-tbl.json" file: `echo {} > data/mock-account-tbl.json`
   - Create the "utils" folder.
   - Create the "utils/create_accounts.py" file: `type nul > "utils/create_accounts.py"`

create_accounts.py

```
import re
import json
import getpass
from datetime import datetime
from uuid import uuid5, NAMESPACE_OID

import bcrypt


USERNAME = re.compile(r"^([a-z]+[a-z\d.]*){1,30}$")
PASSWORD = re.compile(r"^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[!-~]{8,16}$")

def encode_pw(pw):
    return bcrypt.hashpw(pw.encode(), bcrypt.gensalt()).decode()

def ask_password():
    while True:
        pw = getpass.getpass(" Enter password (8-16 chars): ").strip()
        if PASSWORD.fullmatch(pw):
            return pw

def load_accounts():
    with open("data/mock-account-tbl.json") as file:
        return json.load(file)

def update_accounts(accounts):
    with open("data/mock-account-tbl.json", "w", encoding="utf-8") as file:
        json.dump(accounts, file, indent=4)

def main():
    print("\nAccount Generator v1")
    print("\n[GUIDE]\n Hit <ENTER> after an empty username to end program.")
    print(" User level: 0 - Std user, 1 - Moderator, 2 - Admin, 3 - Super Admin")
    print(" Username: lowercase, starts with a-z then (alphanumeric and period).")
    print(" Password: contains at least 1 lowercase, " \
        "1 uppercase, 1 digit, and 1 special character (@$!%*?&)")
    print(" Use the same username to overwrite account.")

    accounts = load_accounts()

    while True:
        print("\n[New Account]")

        uname = input(" Username (1-15 chars): ")
        if not uname:
            break
```


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```
        assert USERNAME.fullmatch(uname), "Invalid username."

        hashed = encode_pw(ask_password())
        user_level = input(" User level: ")[:1]
        full_name = input(" Full Name: ").strip()

        assert user_level in "0123", "Invalid user level."
        assert full_name, "Invalid name."

        account = {}
        account["id"] = str(uuid5(NAMESPACE_OID, uname))
        account["fullName"] = full_name
        account["hash"] = hashed
        account["userLevel"] = int(user_level)
        account["createdAt"] = int(datetime.now().timestamp())

        accounts[uname] = account

    update_accounts(accounts)


if __name__ == "__main__":
    main()
```

i. Create mock accounts for the four types of users.
ii. Remember the passwords well.

- Save the changes.

8. In the same **Command Prompt**, push the changes with the commit message:

Command Prompt

```
git add data
git add utils
git commit -m "[db] Populate mock accounts table"
git push
```

9. In the same **Command Prompt**.
   - Create the "data/mock-content-tbl.json" file.
   - Open the "data/mock-content-tbl.json" file and paste the following contents:

data/mock-content-tbl.json

```
[
    {
        "id": "fc16189b-ddf5-5fea-a36d-d4737ee37bd7",
        "author": "admin",
        "title": "Fox Story",
        "content": "The quick brown fox jumps over the lazy dog.",
        "isSynchronized": 0,
        "createdAt": 1734249472
    }
]
```

   - Save all changes.

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10. In the same **Command Prompt**, push the changes with the commit message:

Command Prompt
```
git add "data/mock-content-tbl.json"
git commit -m "[db] Populate mock contents table"
git push
```

11. In the same **Command Prompt**.
    - Create the "app.py" file.
    - Open the "app.py" file and paste the following contents:

app.py
```
"""
    BleedingHeart-API v0.9.0
    (c) 2025 Bidlisiw Inc.
"""

import json
import hashlib
from datetime import datetime

import bcrypt
from flask import Flask, request, jsonify
from flask_cors import CORS


app = Flask("bleedingheart-api")
CORS(app)


@app.route("/")
def index():
    return set_cors({})

@app.route("/authorize", methods=["POST"])
def authorize():

    payload = request.get_json()
    uname = payload.get("username", "")
    pw = payload.get("password", "")

    full_name = ""
    user_level = -1
    access_token = ""
    message = "Account not found."
    user_level, full_name = authorize_user(uname, pw)
    if 0 <= user_level <= 4:
        access_token = generate_access_token(uname, user_level)
        message = ""
```


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```
    return set_cors({
          "accessToken": access_token,
          "userLevel": user_level,
          "fullName": full_name,
          "message": message,
        })

@app.route("/get_contents", methods=["GET"])
def get_contents():

    payload = request.args
    uname = payload.get("username", "")
    user_level = payload.get("userLevel", -1)
    access_token = payload.get("accessToken", "")

    try:
        user_level = int(user_level)
    except:
        user_level = -1

    expected_token = generate_access_token(uname, user_level)
    if not (0 <= user_level <= 4) or access_token != expected_token:
        return set_cors({"message": "401: Unauthorized"}), 401

    return set_cors({ "contents": load_contents() })

@app.errorhandler(404)
def page_not_found(e):
    return set_cors({"message":"Page not found."})


##################
# utils          #
##################

def authorize_user(uname, pw):
    """ Authenticate user login. """

    # Use actual No/SQL in real projects.
    with open(f"data/mock-account-tbl.json") as file:
        accounts = json.load(file)
        if uname not in accounts:
            return -1, ""

        account = accounts[uname]
        hashed = account["hash"].encode()

        if bcrypt.hashpw(pw.encode(), hashed):
            user_level = account["userLevel"]
```


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```
            full_name = account["fullName"]
            return user_level, full_name

        return -1, ""

def generate_access_token(uname, user_level):
    """ Create a session token. """

    # Use standard approaches (like JWT) in your future projects.
    today = datetime.now().strftime("%Y-%m")
    message = (uname + str(user_level) + today).encode()

    return hashlib.sha1(message).hexdigest()

def load_contents():
    """ Load all contents. """

    # Use actual No/SQL in real projects.
    with open(f"data/mock-content-tbl.json") as file:
        return json.load(file)

def set_cors(payload):
    """ Wraps payload in a JSON response with CORS headers. """

    response = jsonify(payload)
    response.headers.add("Access-Control-Allow-Origin", "*")
    return response
```

- Save the changes.

12. In the same **Command Prompt**, launch the website locally:

Command Prompt

```
flask run
```

- Hold CTRL and click the "http://127.0.0.1:5000" URL to launch in browser.
- Verify that browsers displays an empty JSON response.
- **Screenshot the webpage.**

13. In the same **Command Prompt**.
    - Hit **CTRL + C** to terminate the local server.
    - Create the "vercel.json" file.
    - Open the "vercel.json" file and paste the following contents:

vercel.json

```
{
    "builds": [
        {
            "src": "app.py",
            "use": "@vercel/python"
```


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```
        }
    ],
    "routes": [
        {
            "src": "/(.*)",
            "dest": "app.py"
        }
    ]
}
```

- Save all changes.

14. In the same **Command Prompt**, push the changes with the commit message:

```
Command Prompt
git add app.py
git add vercel.json
git commit -m "[web] Add API routes"
git push
```

15. Go to Vercel (vercel.com/dashboard) website.
    - Click the "**Add new…**" dropdown.
    - Select the "**Project**" option.
    - In the "**Import Git Repository**", search the repository.
        i. Add via **GitHub** if not yet whitelisted.
    - Import and complete the deployment.
        i. Fix and integrate bugs as encountered.
    - Visit the API base URL.
    - **Screenshot the JSON viewer of the browser.**
16. Go back to the **LT6 Dashboard** spreadsheet.
    - Copy the **base URL** of the new API under the backend server.

Access Control

In the previous section, we learning about different types of user levels. When developing a software, young programmers must put into mind that there will be various actors interacting with it in order to use, manage, and maintain it. Simply, once the vendor hands off the software to its client, the management is left to the technical staff as the end-users continually use its services. Depending on the agreement, the maintenance may be performed by the original vendor, a third-party, or in-house maintenance and sustaining team.

Software developers enable access controls at various levels starting from the database, the backend, and in the frontend interface. This ensures that only the right people can access specific sets of data. Generally, the **authentication** module allows the users to be granted privileged access to the system through enrollment or registration, the resulting credentials are then used to authenticate along

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with measures such as standard username-password, 2-factor authentication (2FA), multi-factor authentication (MFA), or other modern cybersecurity safeguards. On the other hand, authorization grants authenticated users to access specific data or features. Its actual implementation varies across different systems.

1. In **Command Prompt**:
    a. Clone the frontend repository locally.
    b. Open the repository in **VS Code** (or Windows Explorer.)
    c. Navigate inside the **components** folder.
    d. Open and modify the "NavBar.vue" file with the following changes.

NavBar.vue

```
<template>
  <client-only>
    <header :class="['bh-navbar', navAdminMode]">
      <nav class="bh-nav-left">
          <a href="/" class="bh-nav-home">
            <img src="/logo.png" width=20 height=20 class="bh-nav-logo">
            <span>Bleedingheart</span>
          </a>
      </nav>
      <nav id="bhNavBasic" class="bh-nav-right bh-hidden">
        <ul>
          <li><a href="#" class="bh-nav-link">3R Guide</a></li>
          <li><a href="#" class="bh-nav-link">Leaderboard</a></li>
          <li><a href="#" class="bh-nav-link">Contribute</a></li>
          <li><span id="bhNavBasicLogout" class="bh-nav-link">Logout</span></li>
        </ul>
      </nav>
      <nav id="bhNavAdmin" class="bh-nav-right bh-hidden">
        <ul>
          <li><a href="#" class="bh-nav-link">Settings</a></li>
          <li><a href="#" class="bh-nav-link">Reports</a></li>
          <li><span id="bhNavAdminLogout" class="bh-nav-link">Logout</span></li>
        </ul>
      </nav>
    </header>
  </client-only>
</template>

<script setup lang="ts">
const props = defineProps<{ navAdminMode: string }>();
</script>
```

    i. Save all changes.

2. Navigate inside the **pages** folder.
    a. Open and modify the "index.vue" file with the following changes.

index.vue

```
<script lang="ts"></script>

<template>
```


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```
  <client-only>
    <NavBar :navAdminMode="navAdminMode" />
    <div id="bhMain" class="bh-main">
      <span id="bhPageTitle" class="bh-page-title bh-hidden">Latest</span>
      <div id="bhContainer" class="bh-container"></div>
    </div>
    <Cookies />
    <Copyright />
  </client-only>
</template>

<script setup lang="ts">
import { useSeoMeta, useHead } from "@vueuse/head";

const title = "Bleedingheart";
const description = "A community-based bin finder and locator application.";

useSeoMeta({
  title: () => title,
  description: () => description,
  charset: "utf-8",
  viewport: "width=device-width, initial-scale=1.0",
});

useHead({
  link: [
    { rel: "icon", type: "image/png", href: "/logo.png" },
    { rel: "stylesheet", href: "/reset.css" },
    { rel: "stylesheet", href: "/custom.css" },
    { rel: "preconnect", href: "https://fonts.googleapis.com" },
    { rel: "preconnect", href: "https://fonts.gstatic.com", crossorigin: "" },
    { rel: "stylesheet", href: "https://fonts.googleapis.com/xxx", },
  ],
});

const globalDelay = 500;

const allowCookies = useCookie<boolean>("allowCookies", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
allowCookies.value = allowCookies.value ?? false;

const retries = useCookie<number>("retries", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60,
});
retries.value = retries.value ?? 0;

const username = useCookie<string>("username", {
  sameSite: "none",
  secure: true,
```


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```
  maxAge: 60 * 60 * 24,
});
username.value = username.value ?? "";

const accessToken = useCookie<string>("accessToken", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
accessToken.value = accessToken.value ?? "";

const userLevel = useCookie<number>("userLevel", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
userLevel.value = userLevel.value ?? -1;

const fullName = useCookie<string>("fullName", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
fullName.value = fullName.value ?? "";

type ContentItem = {
  id: string;
  author: string;
  title: string;
  content: string;
  isSynchronized: number;
  createdAt: number;
};

type ContentResponse = {
  contents: ContentItem[];
};

const router = useRouter();
const bhServer = "https://bleedingheart-api.vercel.app";
let navAdminMode = userLevel.value >= 2 ? "bh-nav-admin" : "";

function getByID<T extends HTMLElement>(id: string) {
  return document.getElementById(id) as T;
}

function checkAuthorizations() {
  if (!accessToken.value) {
    router.push("/login");
  } else if (userLevel.value >= 2) {
    router.push("/admin");
  }

  const navBasic = getByID<HTMLDivElement>("bhNavBasic");
```


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```
  const navAdmin = getByID<HTMLDivElement>("bhNavAdmin");
  const navLogout = getByID<HTMLDivElement>("bhNavBasicLogout");
  const pageTitle = getByID<HTMLSpanElement>("bhPageTitle");

  if (!navBasic || !navAdmin || !navLogout || !pageTitle) return;

  navLogout.addEventListener("click", (e) => {
    retries.value = 0;
    username.value = "";
    accessToken.value = "";
    userLevel.value = -1;
    fullName.value = "";

    router.push("/login");
  });

  navAdmin.remove();
  navBasic.classList.remove("bh-hidden");
  pageTitle.classList.remove("bh-hidden");

  loadModal();
}

function loadModal() {
  const cookieModal = getByID<HTMLDivElement>("bhCookieModal");
  const cookieBox = getByID<HTMLDivElement>("bhCookieBox");
  const cookieX = getByID<HTMLDivElement>("bhCookieX");
  const cookieOK = getByID<HTMLDivElement>("bhCookieOK");
  if (!cookieModal || !cookieBox || !cookieX || !cookieOK) return;

  if (allowCookies.value === true) {
    allowAllCookies();
    return;
  }

  showCookiePopup();

  cookieX.addEventListener("click", () => {
    showCookiePopup(false);
  });

  cookieOK.addEventListener("click", () => {
    allowAllCookies();
  });
}

function showCookiePopup(show: boolean = true) {
  const cookieBox = getByID<HTMLDivElement>("bhCookieBox");
  const cookieModal = getByID<HTMLDivElement>("bhCookieModal");
  if (!cookieBox || !cookieModal) {
    setTimeout(showCookiePopup, 50);
    return;
  }
```


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```
  if (!show) {
    cookieBox.classList.add("bh-hidden");
    cookieModal.classList.add("bh-hidden");
    return;
  }

  cookieBox.classList.remove("bh-hidden");
  cookieModal.classList.remove("bh-hidden");
}

let lastCookieClicked = 0;
function allowAllCookies() {
  if (lastCookieClicked >= Date.now() - globalDelay) return;
  lastCookieClicked = Date.now();
  allowCookies.value = true;
  showCookiePopup(false);
  loadContents();
}

onMounted(() => {
  nextTick(() => {
    checkAuthorizations();
  });
});

async function loadContents() {
  if (!accessToken.value) return;

  var rawContents = [] as ContentItem[];
  try {
    const query = new URLSearchParams({
      username: username.value,
      userLevel: String(userLevel.value),
      accessToken: accessToken.value,
    }).toString();
    const response = (await $fetch(`${bhServer}/get_contents?${query}`, {
      headers: { "Content-Type": "application/json" },
      method: "GET",
    })) as ContentResponse;

    rawContents = response.contents;
  } catch (e: any) {
    return;
  }

  const container = getByID<HTMLDivElement>("bhContainer");
  if (!container) return;

  container.innerHTML = "";
  const contents = renderContents(rawContents) as HTMLElement;
  container.appendChild(contents);
}

function renderContents(data: ContentItem[]): HTMLElement {
```


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```
  const parent = document.createElement("div");
  parent.className = "bh-contents";

  data.forEach((item, index) => {
    const wrapper = document.createElement("div");
    wrapper.id = `content-${index}`;
    wrapper.className = "bh-content-box";

    const title = document.createElement("h2");
    title.className = "bh-content-title";
    title.textContent = item.title;

    const meta = document.createElement("p");
    const date = new Date(item.createdAt * 1000);
    const published = date.toLocaleDateString("en-US", {
      year: "numeric",
      month: "long",
      day: "numeric",
    });
    meta.textContent = `by ${item.author} • ${published}`;
    meta.className = "bh-content-meta";

    const content = document.createElement("p");
    content.className = "bh-content-data";
    content.textContent = item.content;

    wrapper.append(title, meta, content);
    parent.appendChild(wrapper);
  });

  return parent;
}
</script>

<style>
.bh-nav-link:hover {
  cursor: pointer;
}

.bh-page-title {
  position: relative;
  margin: 60px auto 0;
  width: 800px;
  max-width: 90%;
  padding: 20px;
  font-size: 1.8em;
  font-weight: bold;
  border-bottom: 1px solid rgba(0, 0, 0, 0.2);
  display: block;
}

.bh-container {
  position: relative;
  margin: 20px auto 0;
```


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```
  width: 800px;
  max-width: 90%;
}

.bh-content-box {
  margin-bottom: 10px;
  padding: 20px;
  background: #fff;
  border-radius: 3px;
  border: 1px solid rgba(0, 0, 0, 0.2);
  cursor: pointer;
}

.bh-content-title {
  line-height: 1.5em;
  font-weight: bold;
}

.bh-content-meta {
  font-size: 0.8em;
  color: #666;
}

.bh-content-data {
  margin-top: 10px;
  line-height: 1.5em;
}
</style>
```

b. Save all changes.

3. Navigate inside the **pages** folder.
   a. Create the "login.vue" file.
   b. Open and modify the "login.vue" file with the following contents.

login.vue

```
<script lang="ts"></script>

<template>
  <client-only>
    <NavBar :navAdminMode="navAdminMode" />
    <div id="bhMain" class="bh-main">
      <Login />
    </div>
    <Cookies />
    <Copyright />
  </client-only>
</template>

<script setup lang="ts">
import { useSeoMeta, useHead } from "@vueuse/head";

const title = "Bleedingheart";
const description = "A community-based bin finder and locator application.";
```


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```
useSeoMeta({
  title: () => title,
  description: () => description,
  charset: "utf-8",
  viewport: "width=device-width, initial-scale=1.0",
});

useHead({
  link: [
    { rel: "icon", type: "image/png", href: "/logo.png" },
    { rel: "stylesheet", href: "/reset.css" },
    { rel: "stylesheet", href: "/custom.css" },
    { rel: "preconnect", href: "https://fonts.googleapis.com" },
    { rel: "preconnect", href: "https://fonts.gstatic.com", crossorigin: "" },
    { rel: "stylesheet", : "https://fonts.googleapis.com/xxx", },
  ],
});

const globalDelay = 500;

const allowCookies = useCookie<boolean>("allowCookies", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
allowCookies.value = allowCookies.value ?? false;

const retries = useCookie<number>("retries", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60,
});
retries.value = retries.value ?? 0;

const username = useCookie<string>("username", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
username.value = username.value ?? "";

const accessToken = useCookie<string>("accessToken", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
accessToken.value = accessToken.value ?? "";

const userLevel = useCookie<number>("userLevel", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
userLevel.value = userLevel.value ?? -1;
```


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```
const fullName = useCookie<string>("fullName", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
fullName.value = fullName.value ?? "";

const router = useRouter();
let navAdminMode = userLevel.value >= 2 ? "bh-nav-admin" : "";
const bhServer = "https://bleedingheart-api.vercel.app";

type AuthResponse = {
  accessToken: string;
  userLevel: number;
  fullName: string;
  message: string;
};

function getByID<T extends HTMLElement>(id: string) {
  return document.getElementById(id) as T;
}

let lastLoginClicked = 0;
function checkAuthorizations() {
  if (accessToken.value) {
    if (userLevel.value <= 1) {
      router.push("/");
    } else if (userLevel.value >= 2) {
      router.push("/admin");
    }
  }

  const loginBox = getByID<HTMLDivElement>("bhLoginBox");
  const loginForm = getByID<HTMLDivElement>("bhLogInForm");
  const loginUsername = getByID<HTMLInputElement>("bhLoginUsername");
  const loginPassword = getByID<HTMLInputElement>("bhLoginPassword");
  const loginMessage = getByID<HTMLDivElement>("bhLogInMessage");
  const logInActions = getByID<HTMLDivElement>("bhLogInActions");
  const loginOK = getByID<HTMLDivElement>("bhLoginOK");

  if (
    !loginBox ||
    !loginForm ||
    !loginUsername ||
    !loginPassword ||
    !loginMessage ||
    !logInActions ||
    !loginOK
  ) return;

  if (retries.value >= 3) {
    loginForm.classList.add("bh-hidden");
    loginMessage.classList.remove("bh-hidden");
```


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```
    loginMessage.innerText = "Too many attempts, try again after 1 hour.";
    logInActions.classList.add("bh-hidden");
    return;
  }

  loginOK.dataset.busy = "0";
  loginBox.classList.remove("bh-hidden");

  loginOK.addEventListener("click", () => {
    if (loginOK.dataset.busy == "1") return;

    const tempUsername = loginUsername.value;
    const tempPassword = loginPassword.value;

    if (!tempUsername) {
      loginMessage.innerText = "Username is required.";
      loginMessage.classList.remove("bh-hidden");
      return;
    }

    if (!tempPassword) {
      loginMessage.innerText = "Password is required.";
      loginMessage.classList.remove("bh-hidden");
      return;
    }

    authorizeEmail(tempUsername, tempPassword);
  });

  loginUsername.addEventListener("keyup", (e) => {
    if (e.key === "Enter" && loginUsername.value) {
      loginPassword.focus();
    }
  });

  loginPassword.addEventListener("keyup", (e) => {
    if (e.key === "Enter") loginOK.click();
  });

  loadModal();
}

async function authorizeEmail(tempUsername: string, tempPassword: string) {
  if (lastLoginClicked >= Date.now() - globalDelay) return;
  lastLoginClicked = Date.now();

  const loginUsername = getByID<HTMLInputElement>("bhLoginUsername");
  const loginPassword = getByID<HTMLInputElement>("bhLoginPassword");
  const loginMessage = getByID<HTMLDivElement>("bhLogInMessage");
  const loginOK = getByID<HTMLDivElement>("bhLoginOK");

  if (!loginUsername || !loginPassword || !loginOK || !loginMessage) return;

  loginMessage.innerText = "";
```


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```
  loginMessage.classList.add("bh-hidden");
  loginUsername.disabled = true;
  loginPassword.disabled = true;
  loginOK.dataset.busy = "1";

  try {
    userLevel.value = -1;
    const response = (await $fetch(`${bhServer}/authorize`, {
      headers: { "Content-Type": "application/json" },
      method: "POST",
      body: {
        username: tempUsername,
        password: tempPassword,
      },
    })) as AuthResponse;

    accessToken.value = String(response.accessToken);
    userLevel.value = Number(response.userLevel);
    fullName.value = String(response.fullName);
    if (accessToken.value) {
      username.value = tempUsername;
    }
    loginOK.dataset.busy = "0";

    if (!accessToken.value) {
      retries.value = (retries.value || 0) + 1;
    }

    if (response.message) {
      loginMessage.innerText = response.message;
      loginMessage.classList.remove("bh-hidden");
    }
  } catch (e: any) {
    loginMessage.innerText = `Error encountered: ${e}`;
    loginMessage.classList.remove("bh-hidden");
  }

  loginUsername.disabled = false;
  loginPassword.disabled = false;
  loginUsername.value = "";
  loginPassword.value = "";
  loginOK.dataset.busy = "0";
  checkAuthorizations();
}

function loadModal() {
  const cookieModal = getByID<HTMLDivElement>("bhCookieModal");
  const cookieBox = getByID<HTMLDivElement>("bhCookieBox");
  const cookieX = getByID<HTMLDivElement>("bhCookieX");
  const cookieOK = getByID<HTMLDivElement>("bhCookieOK");
  if (!cookieModal || !cookieBox || !cookieX || !cookieOK) return;

  if (allowCookies.value === true) {
    allowAllCookies();
```


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```
    return;
  }

  showCookiePopup();

  cookieX.addEventListener("click", () => {
    showCookiePopup(false);
  });

  cookieOK.addEventListener("click", () => {
    allowAllCookies();
  });
}

function showCookiePopup(show: boolean = true) {
  const cookieBox = getByID<HTMLDivElement>("bhCookieBox");
  const cookieModal = getByID<HTMLDivElement>("bhCookieModal");
  if (!cookieBox || !cookieModal) {
    setTimeout(showCookiePopup, 50);
    return;
  }

  if (!show) {
    cookieBox.classList.add("bh-hidden");
    cookieModal.classList.add("bh-hidden");
    return;
  }

  cookieBox.classList.remove("bh-hidden");
  cookieModal.classList.remove("bh-hidden");
}

let lastCookieClicked = 0;
function allowAllCookies() {
  if (lastCookieClicked >= Date.now() - globalDelay) return;
  lastCookieClicked = Date.now();
  allowCookies.value = true;
  showCookiePopup(false);
}

onMounted(() => {
  nextTick(checkAuthorizations);
});
</script>

<style>
.bh-login-box {
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
  width: 300px;
  padding: 20px;
  background: #fff;
```


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```
  border-radius: 8px;
  border: 1px solid rgba(0, 0, 0, 0.1);
  box-shadow: 0 0 1px #000000bf;
}

.bh-login-header {
  font-size: 1.3em;
  font-weight: bold;
  text-align: center;
  line-height: 2em;
}

.bh-login-slogan {
  margin: 10px 0 20px 0;
  text-align: center;
}

.bh-login-form {
  margin-top: 20px;
}

.bh-login-input {
  margin-bottom: 10px;
  padding: 8px 10px;
  width: 100%;
  font-size: 1.1em;
  border-radius: 3px;
  border: 1px solid rgba(0, 0, 0, 0.3);
  outline: none;
}

.bh-login-message {
  padding: 8px;
  font-size: 0.7em;
  background-color: rgba(255, 0, 0, 0.1);
  border: 1px solid rgba(255, 0, 0, 0.3);
  border-radius: 3px;
}

.bh-login-actions {
  margin-top: 20px;
}

.bh-login-action {
  padding: 10px;
  text-align: center;
  border-radius: 6px;
  border: 1px solid rgba(0, 0, 0, 0.1);
  cursor: pointer;
}

.bh-login-ok {
  color: #fff;
  background-color: #3f7d58;
```


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```
}

.bh-login-ok:hover {
  color: #fff;
  background-color: #3f7d58dd;
}
</style>
```

c. Save all changes.

4. Navigate inside the **pages** folder.
   a. Create the "login.vue" file.
   b. Open and modify the "login.vue" file with the following contents.

login.vue

```
<script lang="ts"></script>

<template>
  <client-only>
    <NavBar :navAdminMode="navAdminMode" />
    <div id="bhMain" class="bh-main">
      <span id="bhSimpleHello" class="bh-simple-hello bh-hidden">
          Hello, {{ fullName }}!
       </span>
    </div>
    <Cookies />
    <Copyright />
  </client-only>
</template>

<script setup lang="ts">
import { useSeoMeta, useHead } from "@vueuse/head";
import { useRouter } from "vue-router";

const title = "Bleedingheart";
const description = "A community-based bin finder and locator application.";

useSeoMeta({
  title: () => title,
  description: () => description,
  charset: "utf-8",
  viewport: "width=device-width, initial-scale=1.0",
});

useHead({
  link: [
    { rel: "icon", type: "image/png", href: "/logo.png" },
    { rel: "stylesheet", href: "/reset.css" },
    { rel: "stylesheet", href: "/custom.css" },
    { rel: "preconnect", href: "https://fonts.googleapis.com" },
    { rel: "preconnect", href: "https://fonts.gstatic.com", crossorigin: "" },
    { rel: "stylesheet", href: "https://fonts.googleapis.com/xxx", },
  ],
});
```


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```
/* Cookies */

const globalDelay = 500;

const allowCookies = useCookie<boolean>("allowCookies", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
allowCookies.value = allowCookies.value ?? false;

const retries = useCookie<number>("retries", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60,
});
retries.value = retries.value ?? 0;

const username = useCookie<string>("username", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
username.value = username.value ?? "";

const accessToken = useCookie<string>("accessToken", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
accessToken.value = accessToken.value ?? "";

const userLevel = useCookie<number>("userLevel", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
userLevel.value = userLevel.value ?? -1;

const fullName = useCookie<string>("fullName", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
fullName.value = fullName.value ?? "";

/* Access Control */

const router = useRouter();
let navAdminMode = userLevel.value >= 2 ? "bh-nav-admin" : "";

function getByID<T extends HTMLElement>(id: string) {
  return document.getElementById(id) as T;
}
```


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```
function checkAuthorizations() {
  if (!accessToken.value) {
    router.push("/login");
  } else if (userLevel.value <= 2) {
    router.push("/");
  }

  const navBasic = getByID<HTMLDivElement>("bhNavBasic");
  const navAdmin = getByID<HTMLDivElement>("bhNavAdmin");
  const navLogout = getByID<HTMLDivElement>("bhNavAdminLogout");
  const simpleHello = getByID<HTMLSpanElement>("bhSimpleHello");

  if (!navBasic || !navAdmin || !navLogout || !simpleHello) return;

  navLogout.addEventListener("click", () => {
    retries.value = 0;
    username.value = "";
    accessToken.value = "";
    userLevel.value = -1;
    fullName.value = "";
    router.push("/login");
  });

  navBasic.remove();
  navAdmin.classList.remove("bh-hidden");
  simpleHello.classList.remove("bh-hidden");

  loadModal();
}

function loadModal() {
  const cookieModal = getByID<HTMLDivElement>("bhCookieModal");
  const cookieBox = getByID<HTMLDivElement>("bhCookieBox");
  const cookieX = getByID<HTMLDivElement>("bhCookieX");
  const cookieOK = getByID<HTMLDivElement>("bhCookieOK");
  if (!cookieModal || !cookieBox || !cookieX || !cookieOK) return;

  if (allowCookies.value === true) {
    allowAllCookies();
    return;
  }

  showCookiePopup();

  cookieX.addEventListener("click", () => {
    showCookiePopup(false);
  });

  cookieOK.addEventListener("click", () => {
    allowAllCookies();
  });
}
```


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```
function showCookiePopup(show: boolean = true) {
  const cookieBox = getByID<HTMLDivElement>("bhCookieBox");
  const cookieModal = getByID<HTMLDivElement>("bhCookieModal");
  if (!cookieBox || !cookieModal) {
    setTimeout(() => showCookiePopup(show), 50);
    return;
  }

  if (!show) {
    cookieBox.classList.add("bh-hidden");
    cookieModal.classList.add("bh-hidden");
    return;
  }

  cookieBox.classList.remove("bh-hidden");
  cookieModal.classList.remove("bh-hidden");
}

let lastCookieClicked = 0;
function allowAllCookies() {
  if (lastCookieClicked >= Date.now() - globalDelay) return;
  lastCookieClicked = Date.now();
  allowCookies.value = true;
  showCookiePopup(false);
}

onMounted(() => {
  nextTick(() => {
    checkAuthorizations();
  });
});
</script>

<style>
.bh-nav-admin {
  background-color: #ec5228 !important;
}

.bh-nav-admin a,
.bh-nav-admin span {
  color: #fff !important;
}

.bh-nav-admin .bh-nav-link:hover {
  background-color: rgba(255, 255, 255, 0.1) !important;
}

.bh-nav-admin .bh-nav-home:hover {
  background-color: rgba(255, 255, 255, 0.1) !important;
}

.bh-nav-admin .bh-nav-link:hover {
  background-color: rgba(255, 255, 255, 0.1) !important;
  cursor: pointer;
```


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```
}

.bh-simple-hello {
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
  width: 300px;
  padding: 20px;
  background: #fff;
  border-radius: 8px;
  border: 1px solid rgba(0, 0, 0, 0.1);
  box-shadow: 0 0 1px #000000bf;
}
</style>
```

   c. Save all changes.

5. Close all opened "*.vue" files.
6. In the **Command Prompt** or integrated terminal:
   a. Install project dependencies: `corepack yarn install`
   b. Install "**prettier**" package: `corepack yarn add prettier`
   c. Install "prettier" package: `corepack yarn prettier`
   d. Prettify all codes: `corepack yarn prettier --write .`
   e. Run the development server: `corepack yarn dev -o`
7. In the launched webpage:
   a. Accept all cookies.
      i. Verify that the login page is visible.
   b. Login using the Standard User account.
      i. Logout account.
      ii. Verify you are redirected to the login page.
   c. Login using the Administrator account.
      i. Logout account.
      ii. Verify you are redirected to the login page.
   d. Fix bugs as encountered.
8. In the same **Command Prompt**, push the changes with the commit message:

Command Prompt

```
git add .
git commit -m "[auth] Add authentication module"
git push
```

9. On the same web browser:
   a. Go to the **frontend** website.
      i. Screenshot the visible webpage.
   b. Accept all cookies.
      i. Verify the login page is visible.

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   ii. **Screenshot the visible webpage.**
c. Login using the Standard User account.
   i. Verify the basic interface is visible.
   ii. **Screenshot the visible webpage.**
   iii. Logout account.
   iv. Verify you are redirected to the login page.
d. Login using the Administrator account.
   i. Verify the administrator interface is visible.
   ii. **Screenshot the visible webpage.**
   iii. Logout account.
   iv. Verify you are redirected to the login page.
e. **Fix bugs and integrate changes as encountered.**
f. Once done, submit it to your instructor via email.

To: **rodney.maniego.jr@vsu.edu.ph**
Subject: [2025-mm-dd] CSci 23 LT6 Surname

Sir Rodney,

Good day!

I am Juan D. Cruz (student #) of your CSci 23 class. Attached are the screenshots for your reference. Below are the links to my GitHub repositories and test websites where I contributed to.

https://github.com/username/codename
https://github.com/username/codename-api
https://codename.vercel.app/
https://codename-api.vercel.app/

Best regards,

Juan D. Cruz
BSCS-2

10. Ensure proper submission for grading; confirm with your instructor if needed.
11. Delete all your files and empty the **Recycle Bin**. (Skip if your own PC)
12. Logout your GitHub, Gmail, Vercel, and VSUEE accounts. (Skip if your own PC)
13. Close the browser, Command Prompt, and other windows.
14. Shutdown the PC and make sure the AVR/UPS is turned off.
15. Arrange the chairs before leaving.

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